PORTI LAB 9

Microsoft Sentinel SIEM & SOAR Implementation

Lab Objective

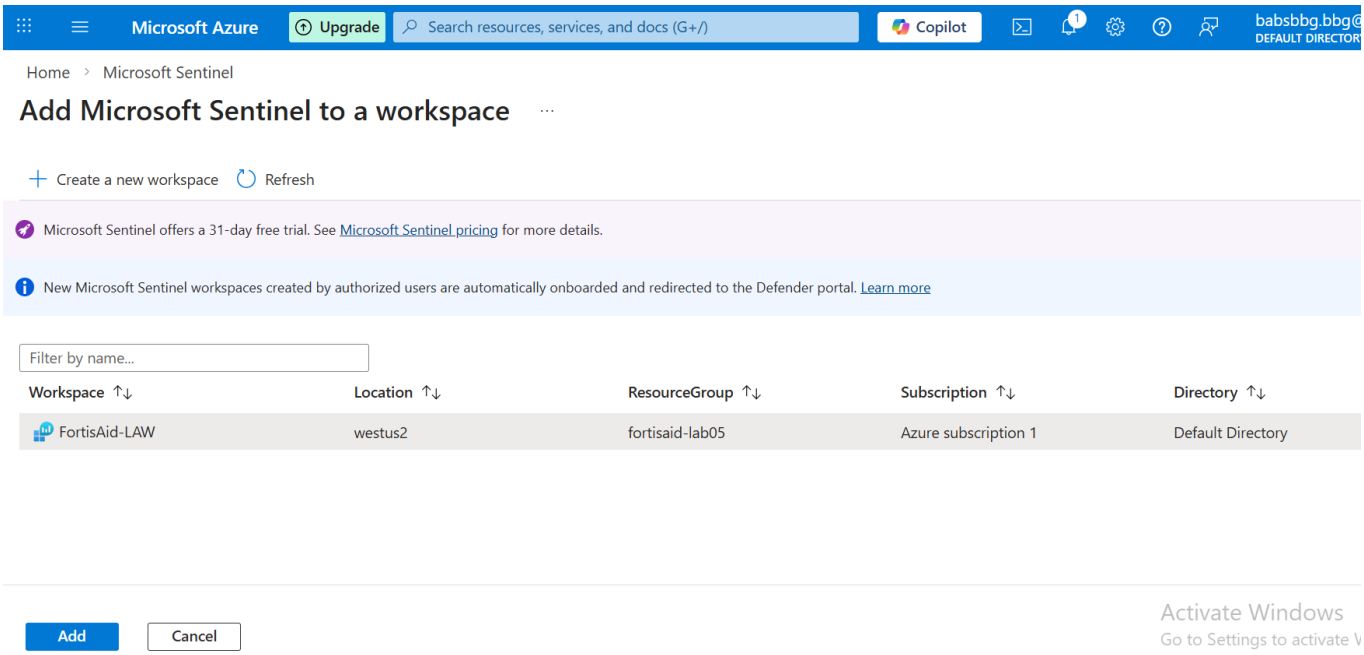
Deploy Microsoft Sentinel on an existing Log Analytics Workspace, ingest Azure Activity logs, create detection rules, and implement a SOAR playbook that automatically escalates incident severity following suspicious administrative activity.

This lab demonstrates an **end-to-end cloud SOC detection and response pipeline**.

Task 1 — Onboard Microsoft Sentinel

Steps

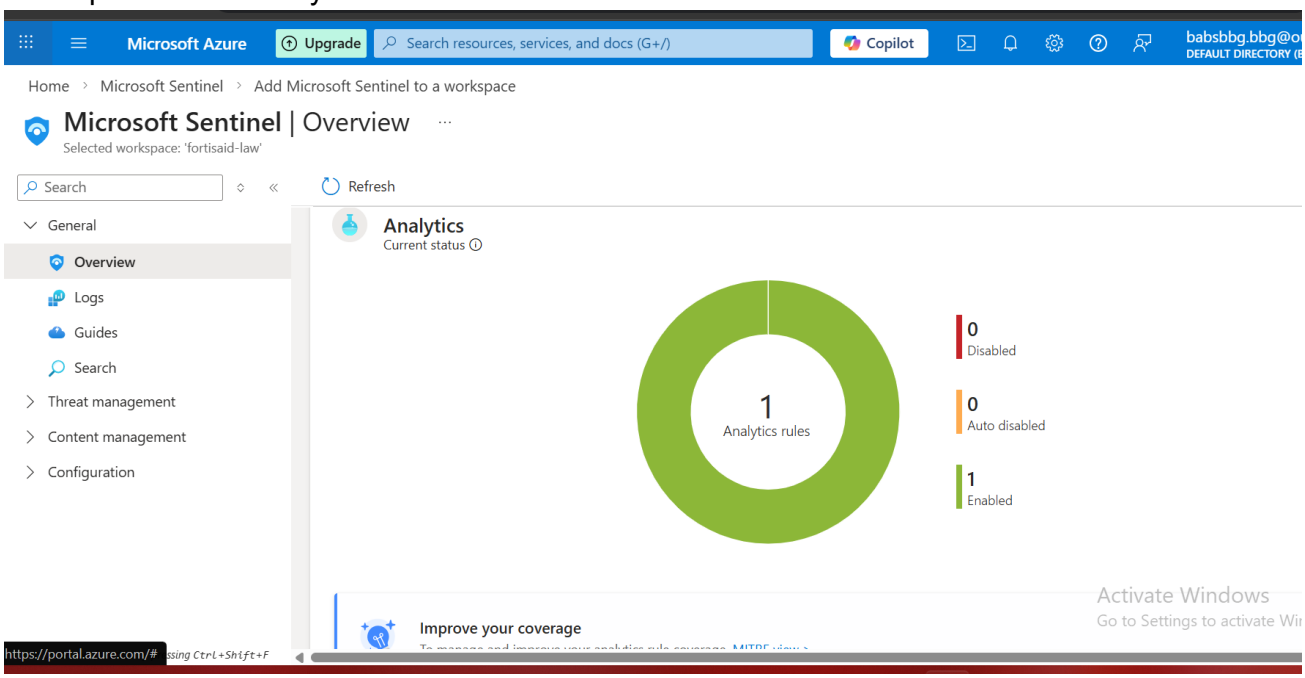
1. Search **Microsoft Sentinel**
2. Click **Create**
3. Select existing LAW in **RG: FortisAid-Lab05**
4. Click **Add**



Sentinel Overview showing workspace successfully onboarded

Verification

- Sentinel Overview loads
- Workspace successfully linked



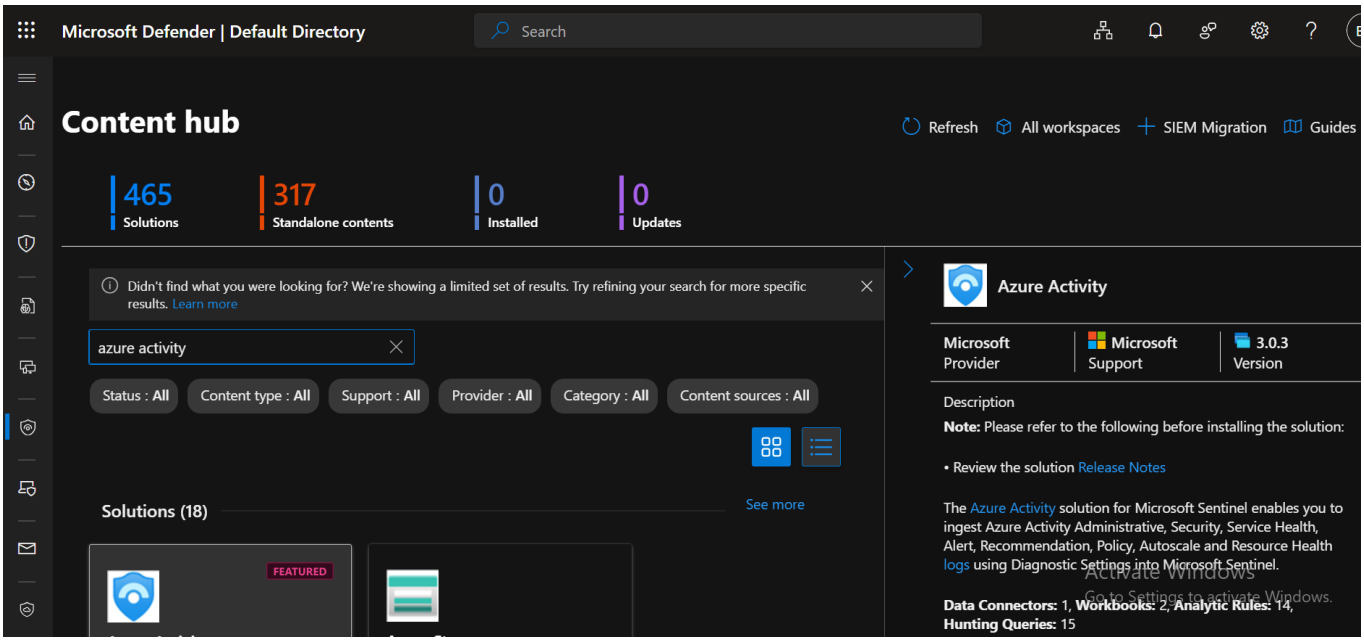
Task 2 — Connect Azure Activity Logs

Steps

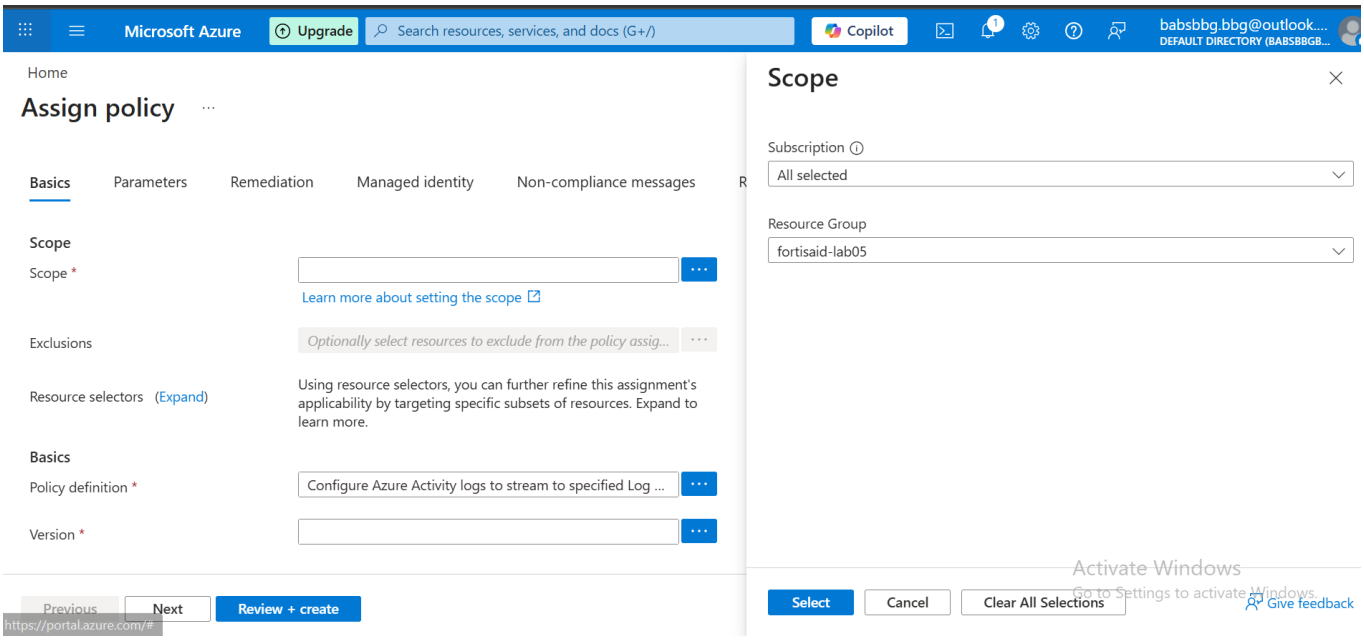
1. Sentinel → **Content hub**
2. Search **Azure Activity** → Install
3. Sentinel → **Data connectors** → Azure Activity → Open
4. Launch **Azure Policy Assignment wizard**
5. Scope = **Subscription**
6. Parameter = Select LAW
7. Enable remediation task
8. Create policy assignment

⚠ Manual diagnostic setting verification performed due to policy propagation delay.

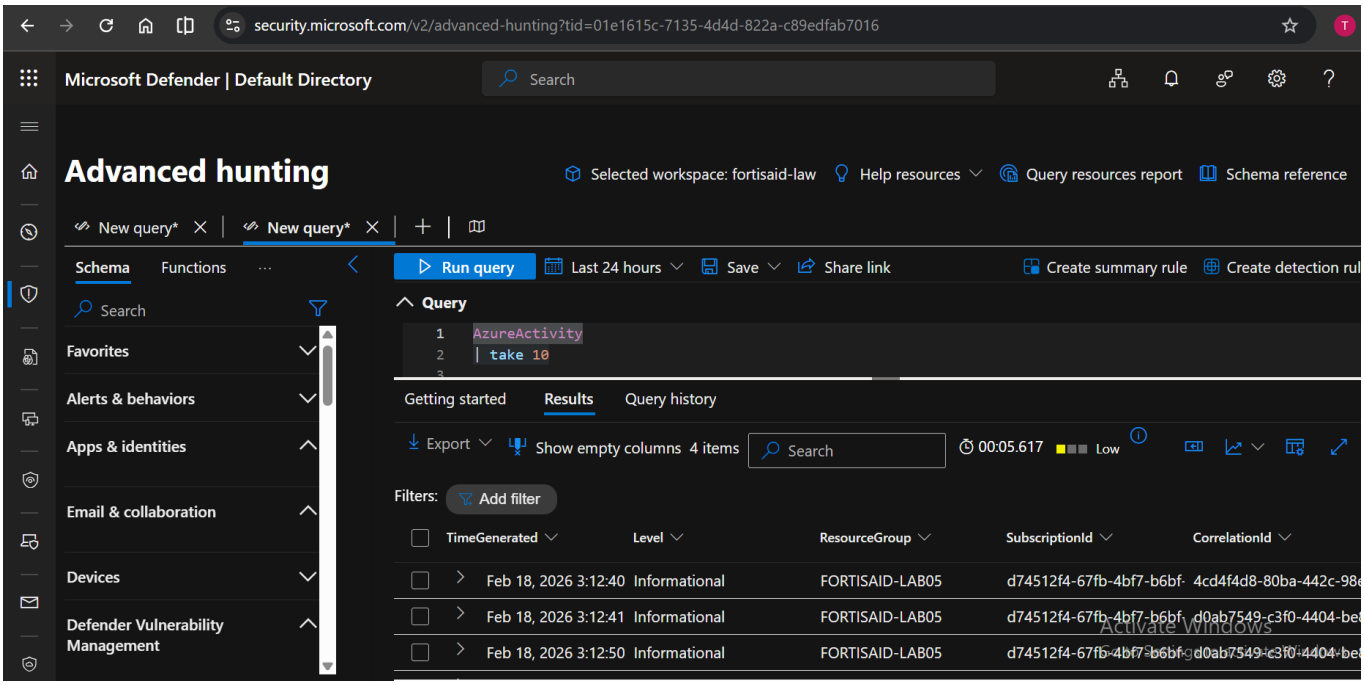
📸 Screenshot Placeholders



Azure Activity connector installed



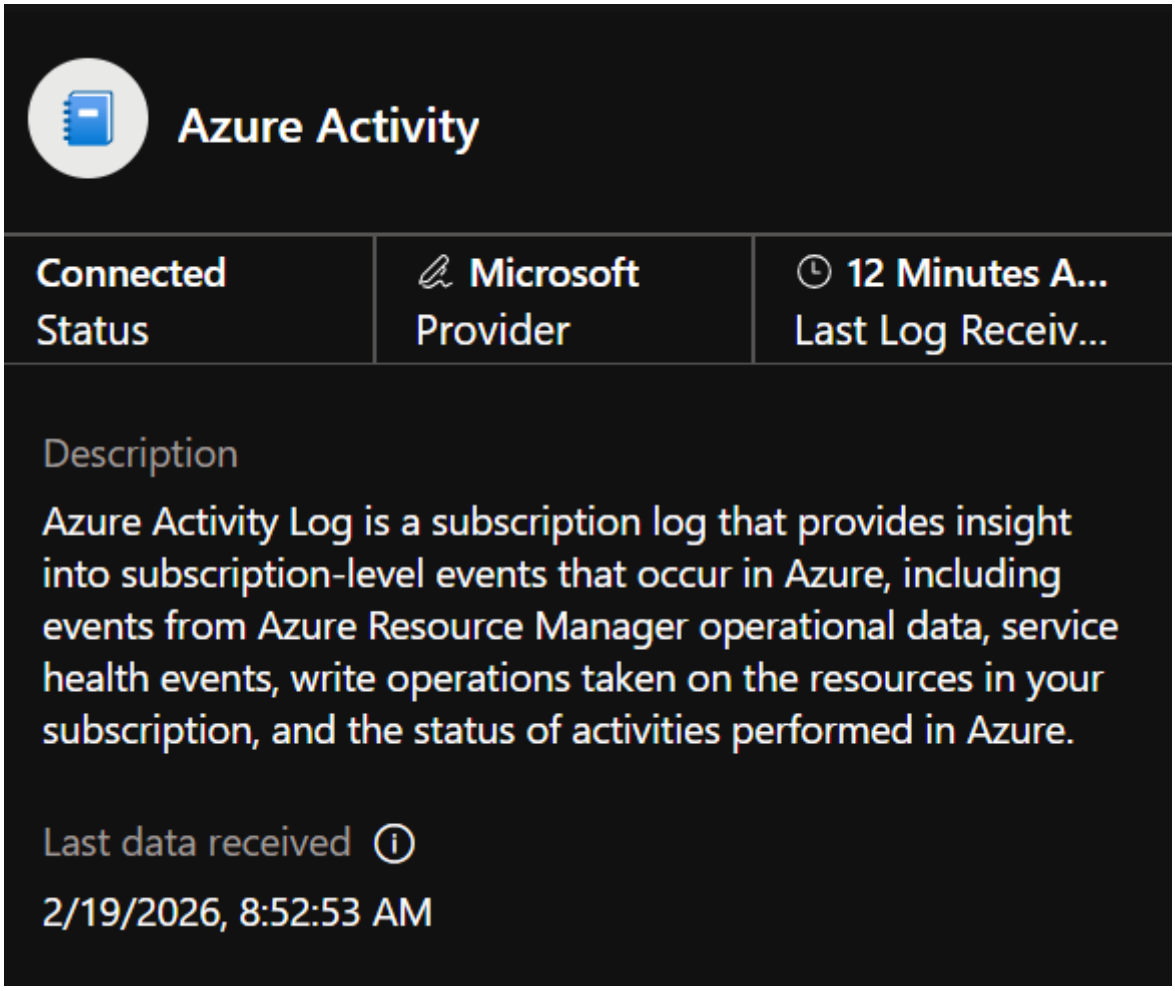
Policy assignment wizard configuration



AzureActivity logs visible in LAW query results

Verification

- AzureActivity table populated

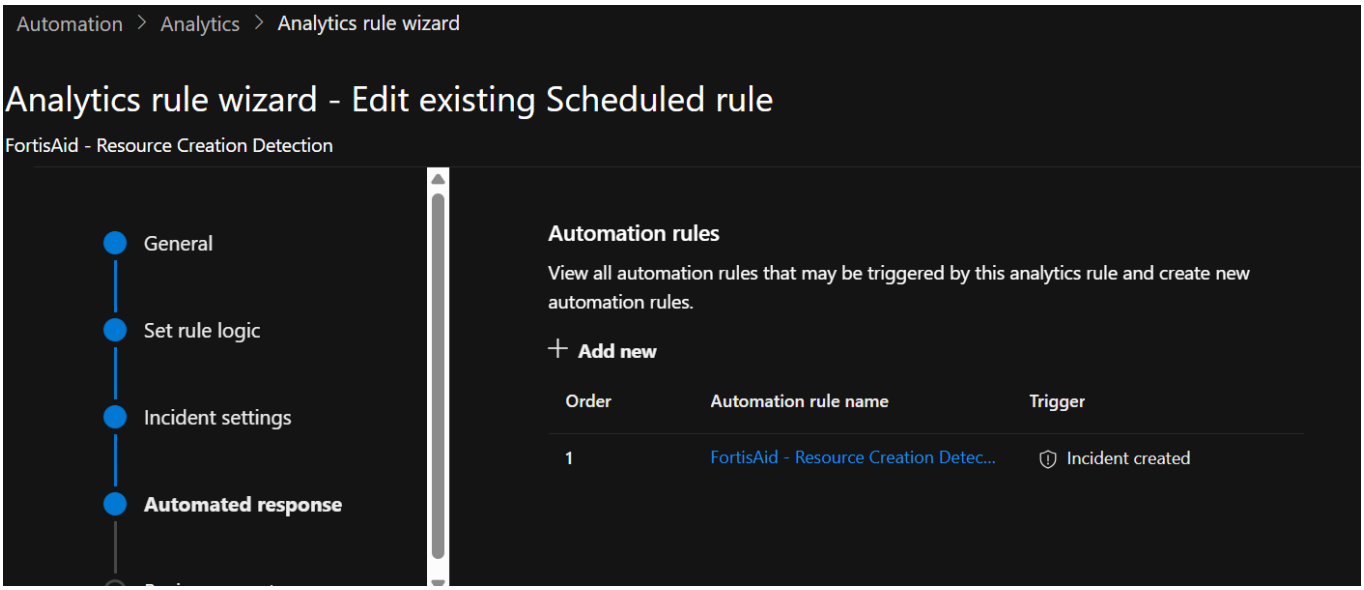


- Connector shows last data received timestamp

Task 3 — Create Detection Rule Using Azure Activity

Steps

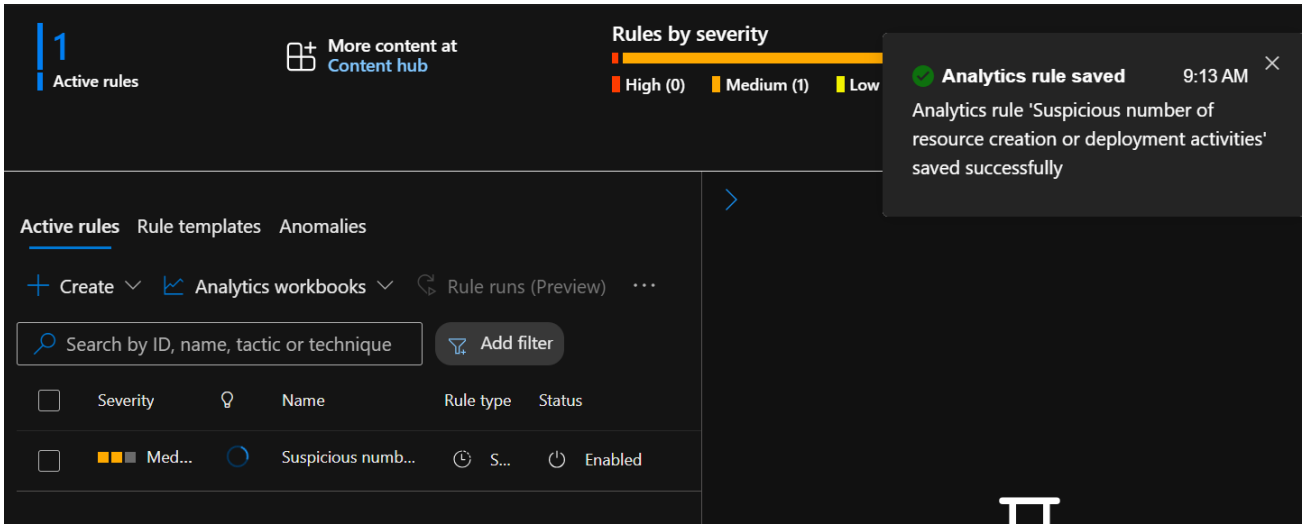
1. Sentinel → **Analytics** → **Rule templates**
2. Deploy **Suspicious number of resource creation or deployment**
3. Save with defaults



Analytics rule active in Sentinel

Verification

- Rule appears in Active Rules

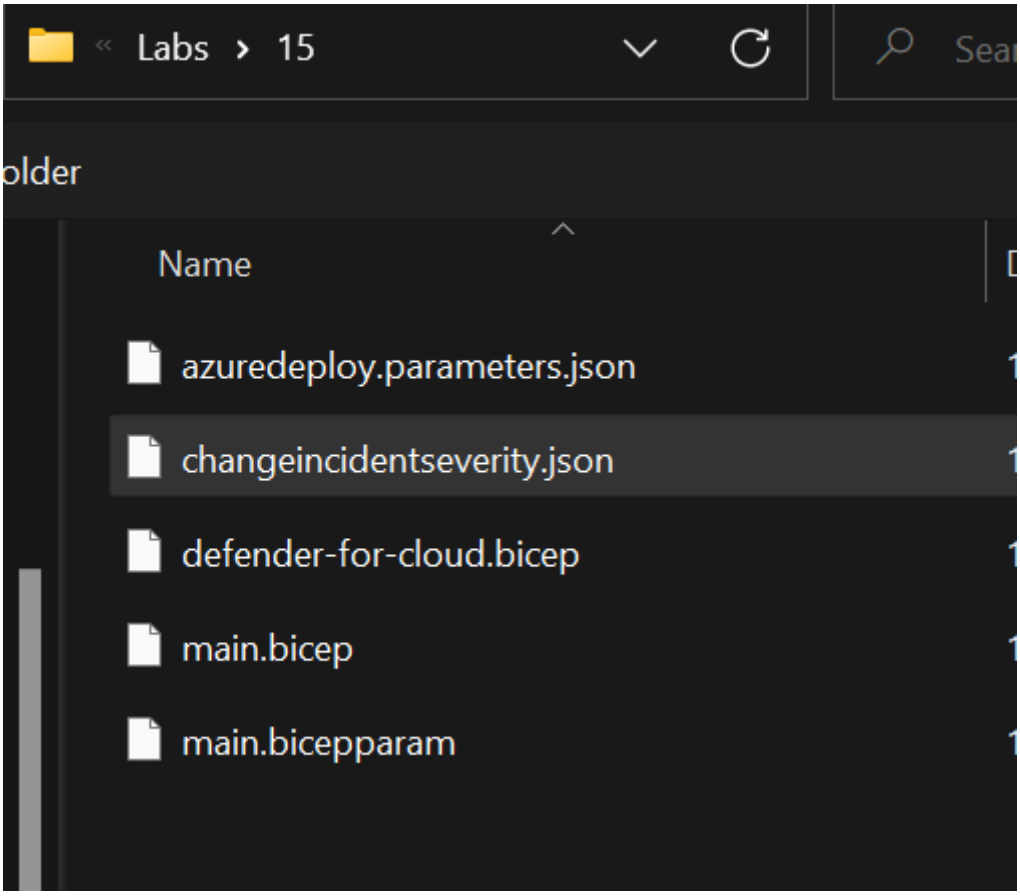


- Incident created during resource creation test

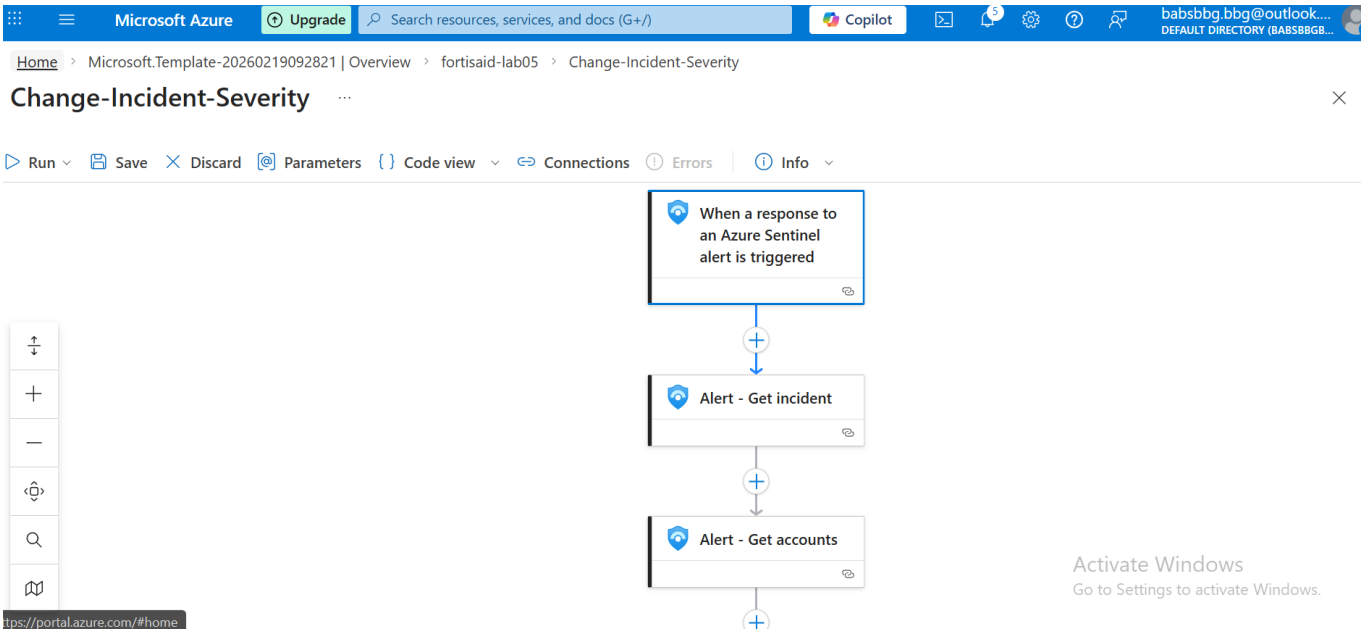
Task 4 — Create SOAR Playbook

Steps

1. Deploy Logic App playbook **Change-Incident-Severity**
2. RG: FortisAid-Lab05
3. Open Logic App → Edit
4. Authenticate Sentinel connectors
5. Add action → **Update Incident**
6. Set severity = High
7. Save



*Logic App playbook deployment



*Logic App designer showing trigger + update incident action

Verification

- Logic App shows 1 trigger + 1 action
- No connection warnings

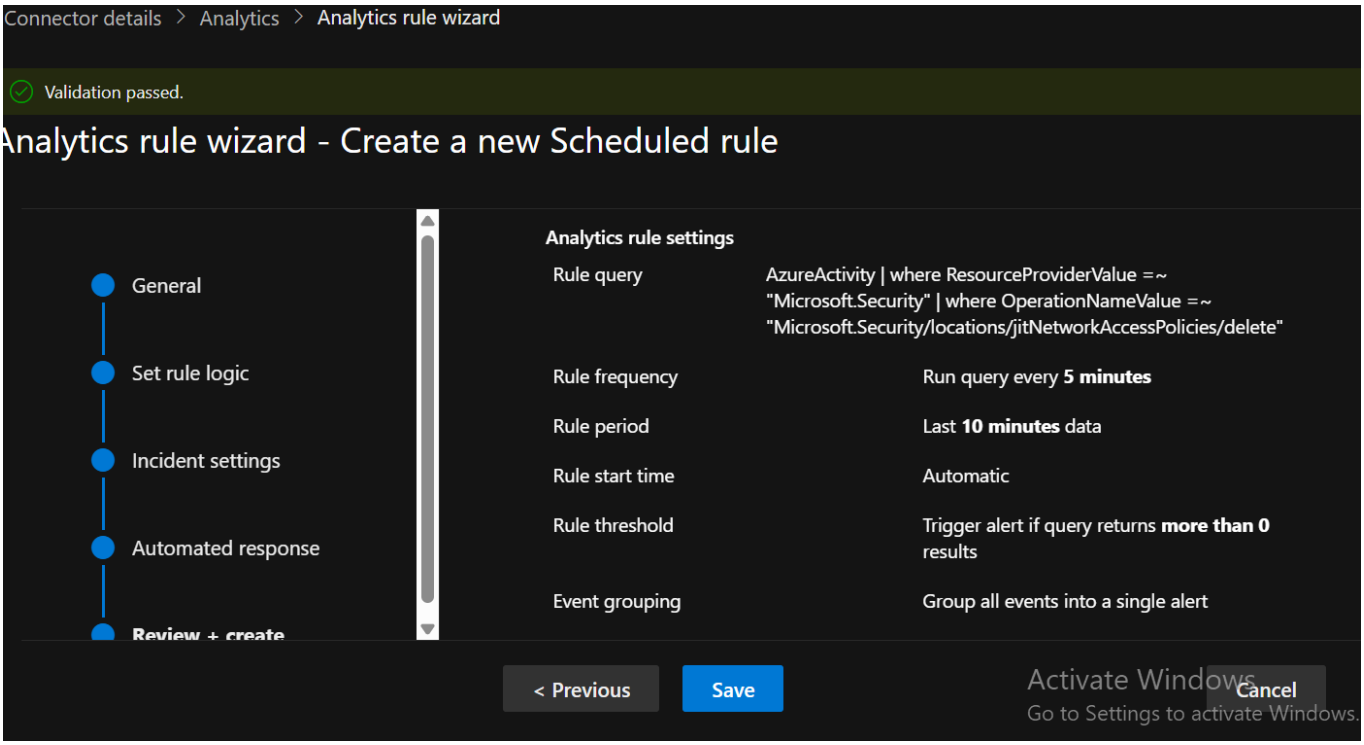
Task 5 — Create Custom Detection + Attach Playbook

Steps

1. Sentinel → Analytics → Scheduled query rule
2. Name: **Playbook Demo**
3. Query:

```
AzureActivity
| where ResourceProviderValue =~ "Microsoft.Security"
| where OperationNameValue =~ "Microsoft.Security/locations/jitNetworkAccessPolicies/delete"
```

4. Frequency = 5 minutes
5. Enable incident creation
6. Attach playbook via automation rule



Custom analytic rule configuration

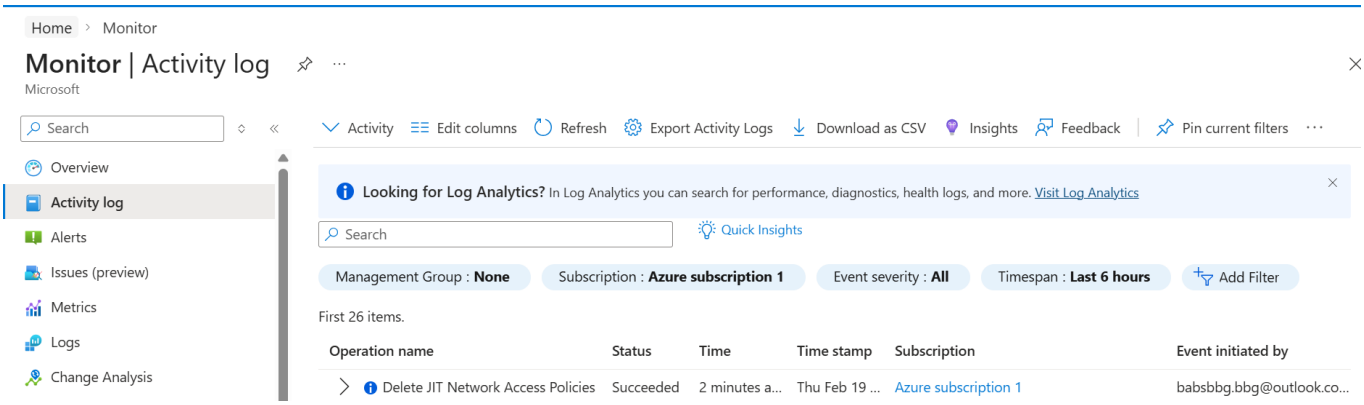
Verification

- Rule enabled
- Automation rule visible
- Permissions configured

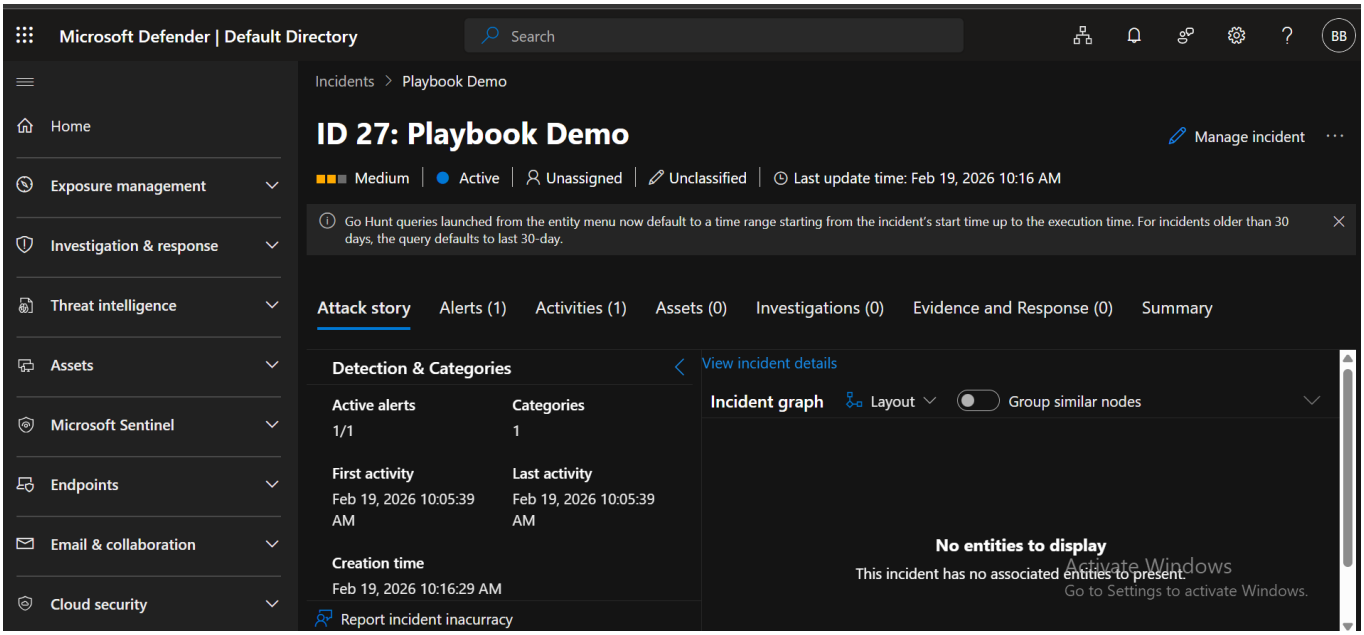
Task 6 — Trigger Incident & Validate SOAR

Steps

1. Defender for Cloud → Remove JIT policy
2. Confirm activity log event
3. Wait for Sentinel rule cycle
4. Open Sentinel → Incidents
5. Confirm playbook run



JIT deletion activity log event

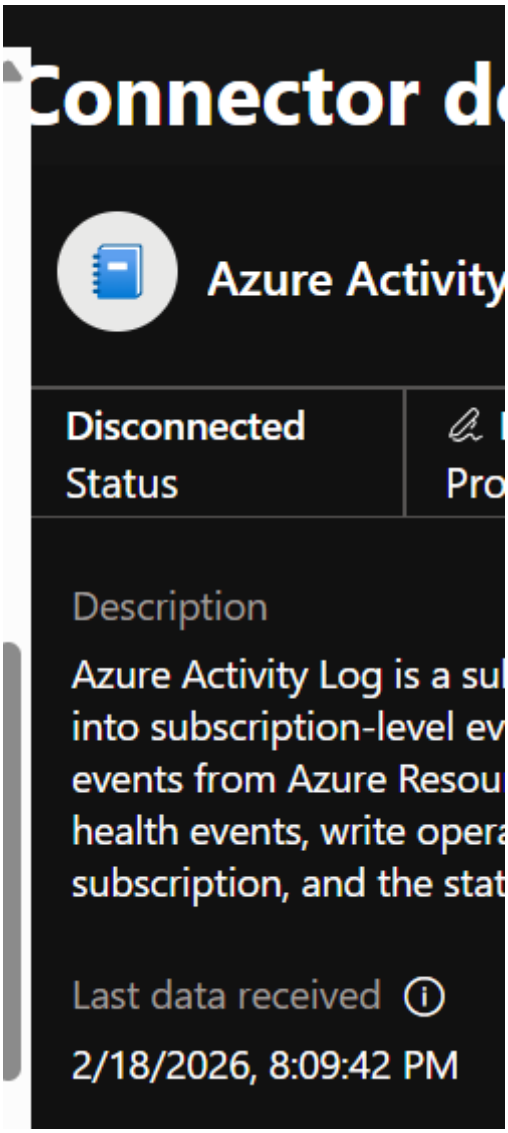


Sentinel incident created

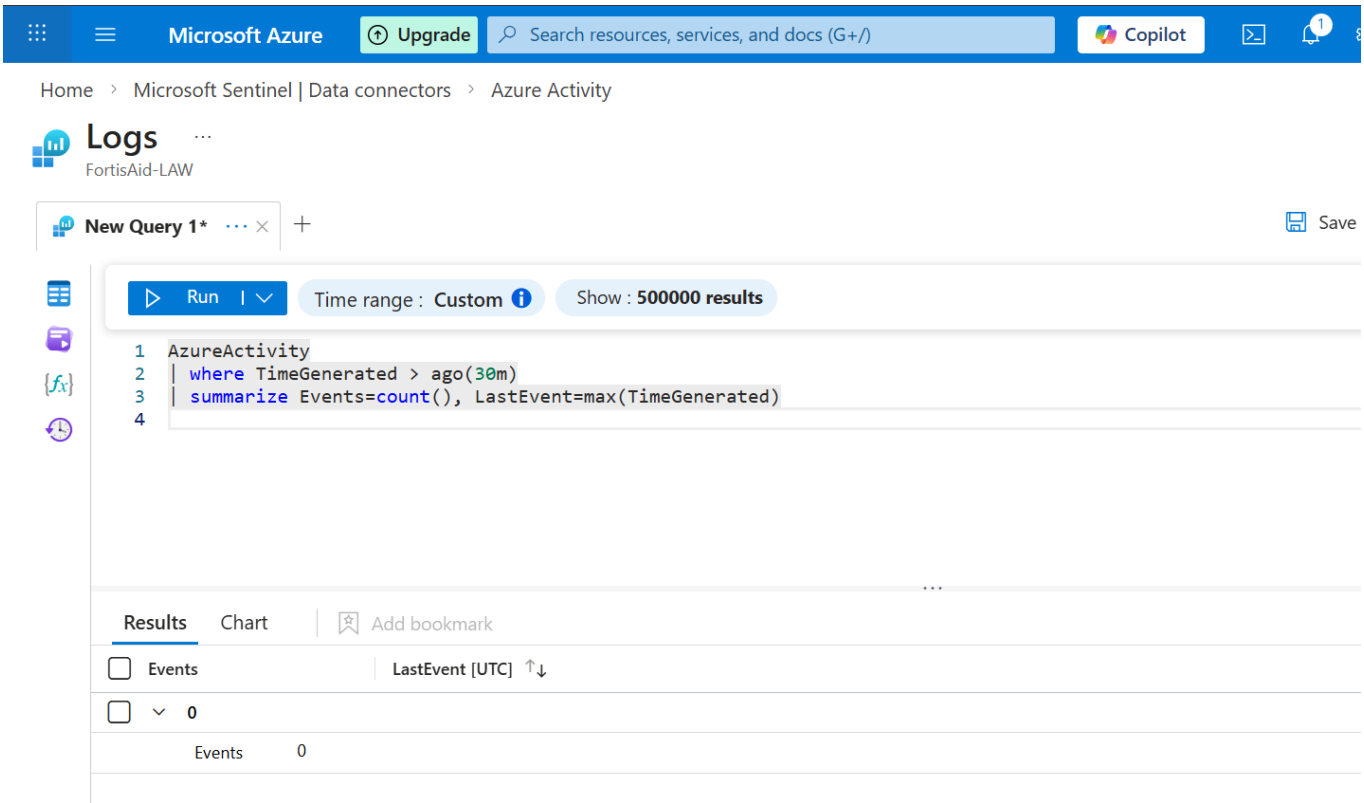
Verification

- Incident created
- Severity escalated automatically
- Logic App run succeeded

Challenges Encountered



Azure Activity connector disconnected state



Analytics rule returning zero results due to ingestion delay

- Azure Policy remediation delay
- Sentinel connector status inconsistency
- Analytics rule ingestion timing mismatch
- RBAC scope blocking playbook execution
- Incident severity not updating until automation configured

Key Technical Learnings

- Azure Activity ingestion delay impacts detection reliability
- Sentinel automation requires subscription-level RBAC
- Incident severity originates from analytic rule
- Playbooks execute only after incident creation
- End-to-end validation requires correlating ingestion → detection → automation

Outcome of the lab

Successfully implemented:

- Sentinel onboarding
- Azure Activity ingestion
- Detection rule deployment
- Incident generation

- SOAR playbook automation
- Automated severity escalation

Demonstrated a complete **cloud SIEM + SOAR detection and response workflow**.